

Industrial Security (Master)

Solution Sheet – Section 07

Safety / Security Co-Engineering

Exercise 7.1 – Mark scheme

Strong plans pair every workshop activity with a deliverable, include both safety and security domain experts plus operations, and name a facilitator who is fluent in both languages.

Exercise 7.2 – Mark scheme

Rows must name the vendor mechanism (e.g. TIA Portal Know-How protection, ControlLogix Source Protection, S7 firewall function-code allowlist) – not abstract “add controls”.

Exercise 7.3 – Sample

- Physical key-switch in RUN, key under dual control.
- Signed program downloads with vendor-rotated certs.
- Continuous SIS code-checksum monitoring (Claroty/Dragos/Nozomi modules).

Exercise 7.4 – Mark scheme

Award marks for: hazard + cyber-hazard in one row, named safety AND security owners, periodic review cadence, link to the SIL verification record so auditors can trace.